

Roll No.:

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B. Tech CSE Cyber Security

Semester: III

Name of the Course: Information Security Foundations

Course Code: TCS 324

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions.
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks.

Q1	(20Marks)	
(a)	Explain the Zero Trust Model. How does it differ from the traditional perimeter-based security approach? Illustrate its core principles with suitable examples.	CO1
(b)	What is access control? Explain different types of access control techniques such as Mandatory Access Control (MAC), and Role-based Access Control (RBAC) with real-world examples.	CO2
(c)	What is CIA Triad. Discuss how confidentiality, integrity, and availability form the pillars of information security. Provide practical examples demonstrating their importance.	CO5
Q2	(20Marks)	
(a)	Why is it important to conduct penetration testing? Explain the various phases of penetration testing with appropriate examples.	CO2
(b)	Write down a shell script program for a port scanner.	CO4
(c)	Are all hackers malicious? Compare white hat, grey hat, and black hat hackers based on their motives, methods, and the legality of their actions.	CO3
Q3	(20Marks)	
(a)	Using the RSA algorithm, perform the encryption on plaintext message $m=7$, where $p=5$, $q=11$ and choose $e=3$. Also calculate the values of public and private key pairs.	CO2
(b)	Write down a python program for the file integrity checking.	CO4
(c)	What is a cryptosystem? Describe the main components of a cryptosystem and explain the working of encryption and decryption processes with a suitable diagram.	CO6
Q4	(20Marks)	
(a)	Differentiate between symmetric key and asymmetric key cryptography. Explain their advantages, disadvantages, and suitable use cases.	CO3
(b)	State and prove how the Double DES algorithm is vulnerable to the meet-in-the-middle attack.	CO5
(c)	Write short notes on the following: a) Data Integrity b) Authentication c) Non-repudiation	CO2
Q5	(20Marks)	
(a)	What is a phishing attack? Explain how phishing works, discuss its types, and describe effective countermeasures to prevent phishing.	CO5

(b)	Explain the working of HTTPS. How does HTTPS ensure confidentiality, integrity, and authentication between a client and a server?	CO6
(c)	Compare and contrast different types of malware attacks, such as keylogger, spyware, viruses, worms, and ransomware. Also, discuss some mitigation strategies for them.	CO3